

Real-Time Detection and Active Mitigation of Command-and-Control Link Exploits in Consumer UAVs

Joseph Park*

Ardsley High School, Ardsley, NY, 10502

Matthew E. Jablonski†

George Mason University, Fairfax, VA, 22030

As consumer Unmanned Aerial Vehicles (UAVs) proliferate in commercial sectors, the security of their Wi-Fi-based command-and-control (C2) links becomes a critical flight safety issue. This study presents the development and live-flight validation of a real-time Intrusion Detection and Response System (IDRS) designed to neutralize unauthorized control attempts. The system was evaluated using the DJI Neo, which relies on Wi-Fi protocols susceptible to interference or hijacking. A two-phase analysis was conducted: first, a vulnerability assessment quantified C2 weaknesses using CVSS v4.0, identifying critical susceptibility to deauthentication and signal injection. Second, a Python-based, rules-driven IDRS was engineered and flight-tested against these vectors. Across 13 live trials, the system achieved a 100% detection rate with a mean response latency of 3.006 seconds, which is sufficiently fast to neutralize threats before flight control is lost. Unlike passive logging systems, this solution features active mitigation, automatically severing unauthorized links to preserve airspace integrity. These results demonstrate that accessible, rules-based logic can effectively harden consumer UAVs against command link exploits without requiring expensive military-grade hardware.

I. Introduction

The commercial and recreational use of small unmanned aerial vehicles (UAVs) has grown rapidly in the past decade, driven by declining hardware costs and advances in flight-control technology. Consumer drones are now deployed across a wide range of sectors, including precision agriculture, infrastructure inspection, cinematography, and emergency response [1]. In more extreme contexts, they have even been configured as improvised weapons in recent conflicts, underscoring both the versatility and the security implications of widely available UAV hardware.

Despite this rapid adoption, security considerations for small UAVs have not kept pace with commercial growth. Manufacturers prioritize production costs, battery endurance, and user accessibility, leaving the protection of wireless communications as an afterthought [2, 3]. Most consumer drones operate on proprietary firmware that lacks the public scrutiny typical of open-source projects, a shortfall compounded by the absence of standardized firmware security guidelines [3, 4]. Compressed product-development cycles further erode security, resulting in a markedly lower standard of protection relative to conventional computing devices [3, 5].

These vulnerabilities are most visible in the Wi-Fi command-and-control (C2) links that consumer drones use to communicate with controllers. Offensive research has shown that unprotected management frames, weak credentials, and insufficient encryption expose those UAVs to hijacking and denial-of-service attacks. Because DJI platforms constitute the vast majority of detected UAS in recent FAA Remote ID traffic analyses [6], vulnerabilities in their specific Wi-Fi stacks carry consequences that could extend to physical hazards such as loss of flight control, property damage, or bystander injury. Despite this risk, defensive research remains sparse. Most proposed intrusion-detection systems either address non-Wi-Fi vectors, rely on offline simulated evaluation, or require specialized hardware.

This study addresses these limitations through a two-phase investigation of the DJI Neo’s Wi-Fi C2 link. Phase 1 conducts a vulnerability analysis through offensive techniques, scoring each finding with the Common Vulnerability Scoring System (CVSS) v4.0 [7]. Phase 2 uses the resulting threat model to engineer a Python-based, rules-driven Intrusion Detection and Response System (IDRS) and evaluates it in live flight against Kismet [8], an industry-standard wireless IDS.

The principal contributions of this work are:

*High School Student Researcher, Ardsley High School, 300 Farm Road, Ardsley, NY, AIAA Student Member, 1925503.

†Assistant Professor, Department of Cyber Security Engineering, 10401 York River Road, MSN 6B1, AIAA Member.

- 1) A CVSS v4.0-scored vulnerability assessment of the DJI Neo’s Wi-Fi C2 link, identifying four exploitable weaknesses and a consolidated two-vector threat model that reduces the defensive problem to deauthentication protection and unauthorized-access prevention.
- 2) A rules-based IDRS with an active mitigation capability that autonomously severs unauthorized connections, moving beyond the passive detection typical of prior work.
- 3) The first live-flight comparative evaluation, to the authors’ knowledge, of a custom Wi-Fi IDRS against an industry-standard system (Kismet) on a consumer UAV, demonstrating matched 100% detection rates while providing active mitigation that the baseline system lacks, using only commodity hardware.

II. Background and Related Work

This section provides background information and related research on the evolution of offensive research against consumer UAVs and the corresponding development of drone-specific intrusion detection systems. Here, we highlight the gaps that motivate our study.

A. Offensive Research on Consumer UAVs

The inadequate standard of security is especially prominent in the Wi-Fi vector of consumer UAVs. Many drones, especially consumer models, are controlled directly by smartphones (e.g., DJI’s “DJI FLY” app and Parrot’s “FreeFlight Mini”). The drone establishes a Wi-Fi network through which the phone and drone can exchange commands, data, telemetry, media, and more [9]. Many of these Wi-Fi-based drone systems lack consistent or even sufficient security measures. Insufficient encryption (such as using Wired Equivalent Privacy or WEP), unprotected management frames, and weak to nonexistent credentials are commonplace among consumer drones [5].

Attacks through the Wi-Fi vector have been thoroughly demonstrated in previous studies. For example, researchers successfully used man-in-the-middle (MITM) and deauthentication attacks to capture private traffic on the Parrot AR 2.0 drone [10]. The accessibility of the equipment used in such attacks, the ease with which they can be conducted, and the high impact that usually results make this vector greatly appealing to attackers. While GPS spoofing and radio-frequency (RF)-based attacks are both equally accessible, Wi-Fi attacks allow a higher level of interaction with the target UAV, such as communication involving sensitive photos, videos, interfaces, and software [3, 11].

These attacks have been thoroughly documented in DJI drones. Researchers have successfully used replay attacks consisting of replicated control packets to hijack and disrupt a DJI Mini SE via Wi-Fi [12]. Schiller et al. [13] identified privacy and security issues in DJI’s RemoteID broadcasts, where the location and sensitive personal information of some operators have been exposed through inadequately protected broadcasts. The same study also exposed 16 different vulnerabilities, including both software- and hardware-based issues, such as arbitrary code execution. Such significant security compromises involving a market leader indicate a widespread neglect of security within the drone industry. While research on the exploitation of Wi-Fi vulnerabilities in UAVs is commonplace, comparatively little work has concentrated on actually designing real-time defense systems for consumer UAVs.

B. Intrusion Detection Approaches for UAVs

The intrusion detection system (IDS), a security concept commonly applied to enterprise-level Wi-Fi networks, has shown promise in defending a variety of systems from external threats, including consumer UAVs [3, 14]. For example, a category of IDS called rules-based systems has been configured with conditional, sometimes adaptive checks that quickly classify threats [14]. These systems cut inference time and computational overhead, which are useful attributes for embedded systems like UAVs [2, 14].

Because rules-based systems can be disadvantaged by their rigidity when faced with dynamic or unknown attacks, anomaly-based, machine-learning models have been integrated into many IDS frameworks [2, 3]. The integration of these models has often augmented IDS performance in different detection contexts. For example, Local Interpretable Model-agnostic Explanations (LIME)-enhanced Support Vector Machine (SVM) models were used to detect intrusions within encrypted UAV Wi-Fi traffic [9]. Generative adversarial networks (GANs) have also been used to generate synthetic adversarial samples, thereby building the model’s resilience against perturbations in GPS spoofing and jamming attacks [15].

Despite these advances, a significant gap persists: many of these defensive frameworks lack live testing and rely on simulated data sourced from public datasets, which may not generalize well to real-world operations. Other studies, such as those by Hassler et al. [16], suggest using cyber-physical features and successfully training models based on

data sourced from live operations. The IDS in that study, though, was evaluated offline, and some studies have conceded limitations regarding the practical accessibility of actually fusing physical sensor data during flight. Thus, despite the presence of numerous vulnerability analyses and threat assessments, the number of proposed defensive frameworks is comparatively few [3, 11]. The majority of such frameworks involve vectors such as GPS spoofing and RF interference, or rely upon simulated model training or evaluation [16, 17], leaving the critical Wi-Fi C2 link largely underserved by real-time, actively mitigating defenses.

III. Study Overview and Experimental Setup

This section details the overall structure of this study in identifying the vulnerabilities (Phase 1) through the development of the IDRS defense solution (Phase 2).

A. Study Overview

This study investigates the Wi-Fi C2 link security of the DJI Neo, a consumer UAV that relies exclusively on Wi-Fi for smartphone-based flight control. The DJI Neo was selected for two reasons: (1) DJI platforms constitute the vast majority of consumer UAS operations, as discussed previously; and (2) the Neo was released in late 2024, shortly before this study began, and no prior security research was publicly available for the model, making it a timely capture of DJI’s current security practices.

Phase 1 subjects the DJI Neo’s Wi-Fi network to four classes of wireless attacks, summarized in Table 1. These attacks were selected because they span the principal IEEE 802.11 threat categories (confidentiality, integrity, and availability) [18, 19] and require only commodity hardware to execute. Each confirmed vulnerability is scored with CVSS v4.0, and the aggregate findings are consolidated into a threat model that identifies the minimum defensive surface.

Table 1 Attack classes tested in Phase 1.

Attack Class	Target Property	Description
Deauthentication	Availability	Spoofed 802.11 management frames force the controller to disconnect from the drone’s Access Point (AP), severing C2 indefinitely.
Address Resolution Protocol (ARP) Spoofing	Confidentiality, Integrity	Falsified ARP replies redirect unicast traffic through the attacker, enabling interception of control data and telemetry.
Denial-of-Service (DoS) / Transmission Control Protocol (TCP) Flood	Availability	High-rate TCP Synchronize (SYN) packets degrade the drone–controller link, increasing control latency.
Service Exploitation	Confidentiality	Unauthenticated access to an HTTP Hypertext Transfer Protocol (HTTP) service (QuickTransfer) retrieves stored images containing GPS metadata.

Phase 2 uses the resulting threat model to build an Intrusion Detection and Response System (IDRS). The IDRS captures 802.11 management frames via `tshark` in monitor mode, preprocesses traffic into 1-second analysis windows, and classifies each window against deterministic detection rules. An active-response module is integrated to autonomously deauthenticate unauthorized devices. To evaluate alternative detection architectures, three machine-learning classifiers (Random Forest, K-Nearest Neighbors, or KNN, and Support Vector Machine, or SVM) are also trained on 1,846 labeled windows of live-captured traffic and evaluated offline via stratified 4-fold cross-validation.

Finally, the rules-based IDRS is evaluated against Kismet [8] in live flight across 13-attack trials (5 deauthentication, 5 unauthorized connection, and 3 combined). Each trial follows a standardized protocol: 30 seconds of pre-trial verification, 30 seconds of attack execution with time-to-alert measured from attack onset, and 30 seconds of recovery. The systems are compared across four primary metrics:

- 1) **Detection rate:** percentage of attacks correctly identified.
- 2) **Detection latency:** time from attack onset to first alert (seconds).
- 3) **Resource overhead:** CPU and memory consumption monitored at 1-second intervals via `pidstat`.
- 4) **Response capability:** ability to actively mitigate detected threats rather than merely logging them.

B. Experimental Setup

All experimentation used industry-standard, open-source tools and software to ensure appropriate rigor. The hardware and software environment described in this section was shared across both phases of the study.

For all wireless testing, a DJI Neo drone was connected over Wi-Fi to an Apple iPhone 14 running DJI’s official “DJI FLY” app on iOS 18.6 to simulate typical, benign operation. All active wireless reconnaissance and attacks were conducted on an ASUS Zenbook 14 OLED (UX3402) running a virtual-machine instance of Kali Linux 2025.2 on VMware Workstation 17 Pro on a Windows 11 host. The Zenbook was equipped with an Alfa AWUS036AXML wireless adapter, connected via USB, for observing the drone’s wireless network in monitor mode. All offensive and defensive tools (e.g., Wireshark, aircmon-ng, aireplay-ng, Nmap, Bettercap, hping3, Burp Suite, and Kismet) were sourced from the Kali Linux package repository. Python 3 with scikit-learn [20] was used for IDS implementation and ML model training. System resource consumption was monitored using the `pidstat` utility at 1-second intervals during live evaluation. Figure 1 shows the test-bench configuration.



Fig. 1 Test-bench configuration: ASUS Zenbook 14 OLED running Kali Linux with Alfa AWUS036AXML wireless adapter (left) and DJI Neo drone (bottom right).

IV. Phase 1: Vulnerability Analysis

This section details the active penetration testing executed against the DJI Neo to identify exploitable Wi-Fi vulnerabilities and derive the operational threat model.

A. Attack Simulation Methodology

The Phase 1 methodology comprised preliminary wireless reconnaissance followed by active penetration testing. First, a baseline analysis of the DJI Neo’s Wi-Fi network was conducted using `nmcli` to identify the Service Set Identifier (SSID), Basic Service Set Identifier (BSSID), and encryption protocols, followed by `nmap` for host discovery and service enumeration. Network traffic was captured for 10–30 minutes using Wireshark [21] to establish a behavioral baseline for the drone’s typical, benign operation.

Following reconnaissance, the four classes of wireless attacks outlined in Section III.A were executed against the drone’s Wi-Fi vector. For each attack class, 3–5 trials were conducted to confirm that the targeted vulnerability was systemic rather than an isolated anomaly.

For the availability attacks, spoofed 802.11 deauthentication packets were generated using `aireplay-ng` and `airmon-ng` [22], explicitly targeting the smartphone controller to force a disconnection from the drone’s network and cause a denial-of-service. Additionally, `hping3` was configured to execute flood attacks against the drone using both high-rate TCP/User Datagram Protocol (UDP) packets and TCP SYN floods.

For confidentiality and integrity attacks, the Bettercap exploitation framework was deployed to spoof ARP responses. This allowed the attack machine to execute a Man-in-the-Middle (MITM) attack, intercepting unicast and ad-hoc peer-to-peer frames exchanged between the DJI Neo and the controller. Finally, open services identified during the `nmap`

scans, specifically port 80/HTTP, were reverse-engineered and probed for unauthenticated access using Burp Suite. All network traffic generated during these attack simulations was captured and saved as .pcap files in Wireshark for subsequent inspection and vulnerability verification.

B. Vulnerability Results and Threat Model

Each confirmed vulnerability was analyzed and quantitatively scored using the CVSS v4.0 framework based on exploitability requirements and operational impact. In total, four exploitable vulnerabilities were identified across the DJI Neo’s Wi-Fi attack surface, summarized in Table 2.

Table 2 CVSS v4.0 vulnerability assessment of DJI Neo Wi-Fi security findings

Vulnerability	Score	Severity	CVSS v4.0 Vector String
Deauthentication Attack	7.1	HIGH	CVSS:4.0/AV:A/AC:L/AT:N/PR:N/UI:N/VC:N/VI:N/VA:H/SC:N/SI:N/SA:N
Service Exploitation	5.9	MEDIUM	CVSS:4.0/AV:A/AC:L/AT:N/PR:L/UI:P/VC:H/VI:N/VA:N/SC:N/SI:N/SA:N
ARP Spoofing	4.6	MEDIUM	CVSS:4.0/AV:A/AC:L/AT:N/PR:L/UI:N/VC:L/VI:N/VA:N/SC:N/SI:N/SA:N
Denial-of-Service (TCP)	3.5	LOW	CVSS:4.0/AV:A/AC:L/AT:N/PR:L/UI:P/VC:N/VI:N/VA:L/SC:N/SI:N/SA:N

Note. Two informational findings, Credential Leakage and Open Port, were identified but had insufficient impacts to be classified as exploitable vulnerabilities.

The **deauthentication attack** was scored as HIGH (CVSS 7.1) due to its severe impact on availability and straightforward exploitability. An attacker requires no authentication and needs only proximity to the drone to execute sustained deauthentication attacks. This can deny Wi-Fi connectivity indefinitely or until critical battery depletion occurs.

The **service exploitation** vulnerability (QuickTransfer) received a MEDIUM severity rating (CVSS 5.9) due to its severe confidentiality impact. Images retrieved via this vulnerability contained GPS metadata revealing sensitive location information about the drone owner, and the images themselves could expose faces, locations, and other personal information.

The **ARP spoofing** vulnerability was also scored as MEDIUM (CVSS 4.6). This attack successfully intercepted unicast traffic between the drone and controller, compromising packets containing control data, firmware information, microcontroller details, and hardware serial numbers. Finally, the **TCP flood** vulnerability received a LOW severity rating (CVSS 3.5), as it degraded service performance and slowed drone operation without causing a complete service disruption.

The critical discovery of Phase 1 emerged from the attack pattern: every MEDIUM- or HIGH-severity vulnerability either directly exploited unprotected management frames (the deauthentication attack) or required prior network authentication with knowledge of the drone’s Wi-Fi password.

This simple pattern yields a consolidated threat model. It dictates that a defensive system needs only to address two fundamental vectors: (1) deauthentication attacks targeting the controlling smartphone, and (2) unauthorized authentication attempts to the drone’s network by unknown Media Access Control (MAC) addresses. By blocking unauthorized network access, the system inherently blocks all authentication-dependent attacks. This simplification came directly from empirical analysis of the drone’s vulnerabilities rather than theoretical assumptions, laying the architectural groundwork for Phase 2.

V. Phase 2: Intrusion Detection and Response System

Based on the empirical threat model, a custom, Python-based Intrusion Detection and Response System (IDRS) was developed to monitor the DJI drone’s network traffic via a wireless adapter in monitor mode. The objective was to design

a lightweight system capable of active mitigation, outperforming or matching the capabilities of industry-standard tools like Kismet.

A. System Architecture and Active Mitigation

The IDRS was designed strictly around the deterministic threat model identified in Phase 1. It relies on a rules-based detection layer implemented via Python scripts that spawn the `tshark` [21] packet-capture utility. The sniffer module isolates 802.11 management frames targeting the drone's specific BSSID, extracting frame type, subtype, source/destination MAC addresses, and signal strength.

Captured traffic is preprocessed into 1-second analysis windows. For each window, the preprocessing module extracts specific operational features: the number of deauthentication frames, the number of authentication or association requests from unknown MAC addresses, the number of EAPOL frames from unknown addresses, and the percentage of overall traffic originating from unauthorized devices. Within each window, the classification module evaluates these features against two primary rule categories:

- **Deauthentication Detection:** Any deauthentication frame directed at a registered MAC address (the drone or the authorized controller) immediately triggers an alert.
- **Unauthorized Connection Detection:** Any connection attempt (authentication requests, association/reassociation requests, or Extensible Authentication Protocol over LAN (EAPOL) handshakes) originating from an unregistered MAC address is flagged as an intrusion.

A critical differentiator of this custom IDRS is its *Active Mitigation Mechanism*. Because the drone's C2 link relies on a point-to-point network topology, any unknown MAC address attempting authentication is inherently malicious. When the IDRS detects an unauthorized connection attempt, the active response module autonomously deploys `aireplay-ng` to broadcast spoofed deauthentication frames to the attacking MAC address. This forcefully severs the attacker's connection before a Wi-Fi Protected Access 2 (WPA2) handshake can be completed, effectively neutralizing all authentication-dependent vulnerabilities (such as service exploitation and ARP spoofing) identified in Phase 1 without risking disconnection of the legitimate controller.

To support operational deployment, the IDRS incorporates a dedicated alert engine and user interface that displays real-time threat notifications alongside session statistics. To prevent alert fatigue during sustained intrusions, the engine maintains a continuous alert state governed by a configurable cooldown period. Upon the conclusion of a flight session, the system automatically compiles and saves a comprehensive log of capture statistics and threat details to the host system for post-flight forensic analysis.

B. Machine Learning Evaluation and Pivot

The initial experimental design proposed a hybrid architecture combining the rules-based layer with a machine learning (ML) classifier to handle ambiguous traffic. To evaluate this, 1,846 1-second windows of live-captured drone traffic (852 benign, 994 malicious) were labeled. Three classifiers, Random Forest, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM), were trained on metadata features (e.g., packet rates, frame counts) and evaluated using stratified 4-fold cross-validation.

Offline evaluation yielded excellent results: Random Forest achieved 98.81% accuracy, KNN achieved 95.07%, and SVM achieved 92.63%. Despite this strong offline performance, preliminary attempts to integrate these models into the real-time IDRS introduced heavy latency as well as computational overhead. Beyond these latency issues, the Phase 1 vulnerability analysis confirmed that the threat model was deterministic rather than probabilistic. The rules-based system was fully capable of classifying malicious frames with 100% certainty based on MAC registration alone. As a result, the ML layer was discarded in favor of the strictly rules-based architecture, eliminating computational overhead while maintaining optimal detection accuracy.

C. Live Evaluation Methodology

The rules-based IDRS was evaluated against Kismet, an open-source, industry-standard wireless IDS, in a controlled, live-flight environment. Both systems were executed concurrently on the Kali Linux attack machine, locked to the drone's operating channel (Ch. 149).

The evaluation comprised 13 live attack trials distributed across three scenarios:

- 1) **Deauthentication Attack:** 5 trials targeting the authorized controller.
- 2) **Unauthorized Connection:** 5 trials initiating connections via `wpa_supplicant`.

3) **Combined Attack:** 3 trials executing both vectors simultaneously.

Each trial followed a standardized 90-second protocol: 30 seconds of pre-trial benign verification (confirming stable flight and controller connection), 30 seconds of active attack execution (with time-to-alert measured via synchronized stopwatch), and 30 seconds of recovery. For each trial, the system’s detection outcome, time-to-first-alert, and the drone’s physical behavioral response were recorded to ensure active mitigation did not induce flight instability. System resource consumption (CPU and memory usage) was monitored continuously at 1-second intervals using the `pidstat` utility.

D. Detection Accuracy and Latency Results

Both the custom IDRS and Kismet achieved perfect detection accuracy across all attack scenarios in live testing. Table 3 presents the detection accuracy results for each attack type.

Table 3 Detection accuracy comparison between custom IDRS and Kismet

Attack Type	Trials	IDRS Accuracy (%)	Kismet Accuracy (%)	IDRS Mitigation Rate (%)
Unauth. Connection	5	100.00	100.00	100.00
Deauthentication	5	100.00	100.00	N/A
Combined Attack	3	100.00	100.00	100.00
Overall	13	100.00	100.00	N/A

Note. Both systems successfully detected all 13 attack trials with no false negatives. The IDRS Mitigation rate metric only applied to attempted unauthorized Wi-Fi connections.

Both intrusion detection systems successfully detected all 13 attack trials with zero false negatives, and neither system generated false-positive alerts during the benign operational periods. A major distinction, however, was the custom IDRS successfully deploying its active mitigation mechanism, automatically deauthenticating the attacking host in all 8 connection-based trials. Kismet, as a passive IDS, could only alert on these intrusions but could not mitigate them.

While detection accuracy was identical, the systems differed in detection latency, summarized in Table 4.

Table 4 Detection latency comparison across attack types

Attack Type	Trials	IDRS Mean (s)	IDRS SD (s)	Kismet Mean (s)	Kismet SD (s)
Unauth. Connection	5	2.214	0.78	1.200	0.34
Deauthentication	5	2.370	0.79	1.800	0.51
Combined Attack	3	5.385	2.93	3.000	3.24
Overall Mean	13	3.006	—	1.846	—

Kismet demonstrated faster detection latency across all attack scenarios, achieving an overall mean latency of 1.846 seconds compared to the custom IDRS’s 3.006 seconds. For single-vector attacks, the IDRS exhibited consistent latency ($SD \leq 0.79$ s across 10 trials), with no trial exceeding 3.4 seconds. The latency gap was most pronounced during combined attacks; however, both systems exhibited high variance under this scenario (IDRS $SD = 2.93$ s, Kismet $SD = 3.24$ s across only 3 trials), suggesting this variability reflects the inherent unpredictability of simultaneous dual-vector attacks rather than a systematic weakness of the IDRS. The IDRS’s 3.006-second mean latency remains operationally acceptable. Because a full unauthorized authentication handshake (probe, authentication, and association exchanges) typically requires 3–5 seconds even under ideal conditions, the IDRS is still capable of triggering its active response fast enough to deny network access before post-authentication attacks can be executed.

E. Resource Utilization

System resource consumption was monitored continuously throughout all trials using `pidstat` at 1-second intervals. Table 5 presents the CPU and memory utilization statistics for both systems.

The custom IDRS operated with a lower mean CPU utilization (21.29%) compared to Kismet (32.45%). As visualized in Figure 2, the IDRS maintained stable processing overhead during active attack periods. Kismet exhibited

Table 5 CPU and Memory utilization statistics

Metric	CPU Usage (%)		Memory Usage (%)	
	IDRS	Kismet	IDRS	Kismet
Mean	21.29	32.45	5.90	1.50
Median	26.00	22.00	6.06	1.75
95th Percentile	34.00	82.34	6.19	2.07
Maximum	46.00	124.00	6.25	2.13

severe computational spikes under duress, particularly between seconds 80–320, reaching a 95th percentile CPU usage of 82.34% and a maximum spike of 124.00%, whereas the IDRS peaked at only 46.00%.

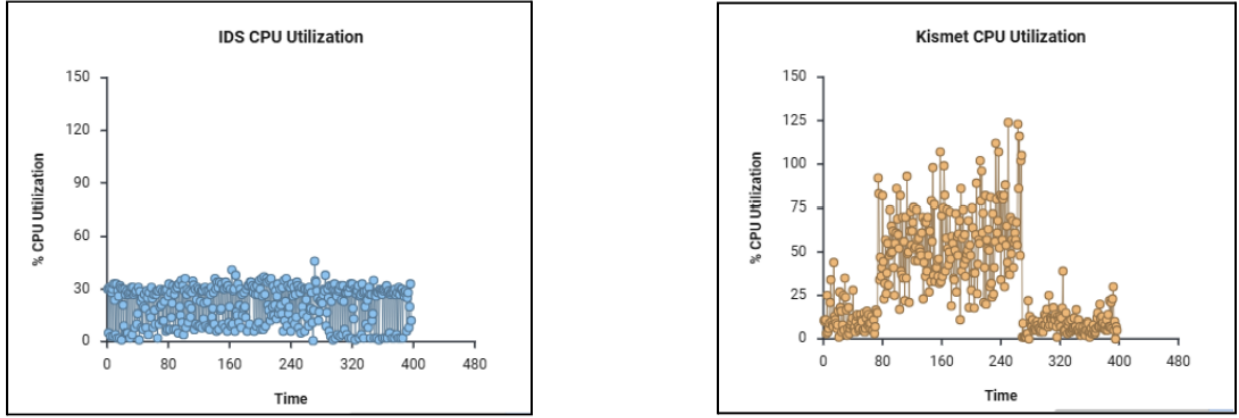


Fig. 2 CPU Utilization over Time. (A) Custom IDRS. (B) Kismet.

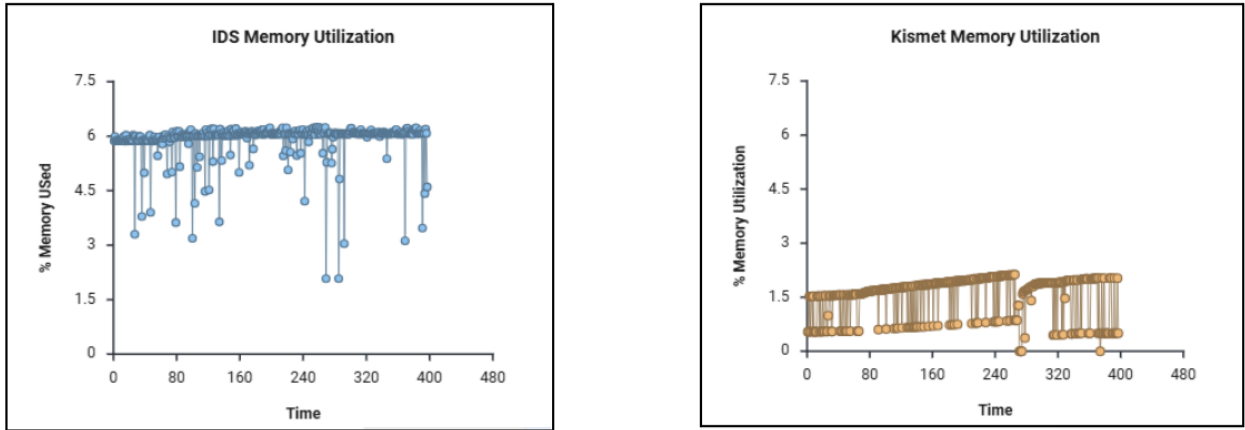


Fig. 3 Memory Utilization over Time. (A) Custom IDRS. (B) Kismet.

As shown in Figure 3, both systems maintained relatively stable memory consumption. While the IDRS consumed roughly four times more memory on average (5.90% vs. 1.50%), this footprint is negligible on modern hardware and remains well within acceptable limits for the host system. The significantly lower and more stable CPU footprint of the IDRS makes it highly suitable for embedded deployment on resource-constrained UAV hardware, where CPU efficiency directly impacts battery endurance.

VI. Discussion

This section contextualizes the empirical findings from both phases of the study, interpreting the DJI Neo’s vulnerability hierarchy to evaluate the operational viability and broader security implications of the custom IDRS.

A. Vulnerability Hierarchy and the Threat Model

The vulnerability analysis of the DJI Neo revealed a troubling security hierarchy: the only HIGH-severity vulnerability requires no network authentication, while all MEDIUM-severity vulnerabilities require prior network access. This pattern suggests that the drone’s primary security boundary, preventing unauthorized network access, is weaker than its internal network mechanisms. Once an attacker manages to bypass the WPA2 handshake, they encounter multiple exploitable services, such as the QuickTransfer feature leaking GPS metadata or unicast traffic vulnerable to ARP spoofing. More concerning is that the deauthentication vulnerability bypasses the handshake entirely because the drone lacks 802.11w management frame protection. As a result, an unauthenticated attacker can persistently sever the drone-controller communication within 3 to 10 seconds. This execution window is particularly critical because it allows an attacker to cut off flight control faster than a user can realistically identify the source of the interference. The persistence of unprotected management frames in a 2024 consumer UAV release, despite standardized IEEE solutions existing since 2009, exemplifies an industry-wide pattern of prioritizing production costs over cyber-physical security.

Despite this deficient security posture, the specific attack patterns directly informed a highly effective, deterministic threat model. Every medium- or high-severity vulnerability either directly exploited unprotected management frames or required prior network authentication. Consequently, the defensive scope could be reduced to two fundamental vectors: addressing deauthentication frames and blocking unauthorized MAC addresses. By preventing unauthorized network access entirely, the system inherently neutralizes all subsequent authentication-dependent attacks without needing the computational overhead or probabilistic uncertainty of a machine learning layer.

B. Detection Performance and System Trade-offs

In live testing, the custom rules-based IDRS maintained specificity without sacrificing sensitivity, achieving perfect detection accuracy with zero false positives during benign operations. While Kismet demonstrated a faster mean detection latency across all attack scenarios (1.846 seconds versus 3.006 seconds), the custom IDRS compensates for this delay with its active mitigation capability. The latency gap is operationally acceptable when considered against the standard timeline of an attack. While the manual timing of the attack onset introduces a potential measurement error of approximately ± 100 –500 milliseconds, this variance does not affect the qualitative conclusion regarding operationally acceptable performance. Unauthorized connection attempts require multiple frame exchanges, specifically probe, authentication, and association requests and responses, before network access is granted. Because this handshake typically takes 3 to 5 seconds even under ideal conditions, the IDRS is capable of triggering its automated `aireplay-ng` response fast enough to deny network access before an attacker can begin executing post-authentication exploits. While the custom system falls short of Kismet’s detection speed, it provides an active response capability that standard passive IDSs lack entirely.

The resource consumption profile of the IDRS further supports its viability for UAV integration. The system operated with a substantially lower and more stable CPU utilization than Kismet, which exhibited severe computational spikes reaching up to 124% maximum usage during active attacks. Although the IDRS consumed roughly four times more memory than Kismet, this footprint remains negligible on modern hardware. Because battery life is one of the strictest constraints in drone technology, the lower CPU overhead of the IDRS makes it highly optimized for embedded deployments aboard resource-constrained UAV hardware.

C. Limitations and Future Work

As this study focused exclusively on the DJI Neo within a controlled laboratory environment, the findings may not generalize to other drone platforms with different Wi-Fi implementations or proprietary firmware. A more comparative evaluation across multiple DJI models is necessary to measure generalizability and identify model-specific vulnerabilities, while field testing in realistic environments would establish system robustness beyond laboratory conditions. The evaluation did not assess highly advanced adversaries employing anti-detection techniques such as MAC address randomization, rate-limited probing, or attacks executed from maximum signal range under degraded RF conditions.

Future development of the IDRS should prioritize expanding its operational capabilities for fleet-oriented deployments. The current system monitors a single BSSID and requires manual MAC address registration. Integrating automated

configuration mechanisms and multi-drone monitoring would significantly improve its utility. The system’s detection logic could also be expanded to address control-packet replay attacks and Remote ID broadcast exploitation to provide comprehensive protection across the full spectrum of documented DJI vulnerabilities. Future iterations must investigate the regulatory compliance of autonomous deauthentication mechanisms and their potential to cause unintended interference with adjacent, benign Wi-Fi networks.

VII. Conclusion

As consumer UAVs are increasingly integrated into sensitive sectors, the persistent vulnerability of their Wi-Fi C2 links poses a critical operational hazard. This study demonstrates that effective, active protection for these networks is achievable without relying on computationally expensive machine learning models or specialized hardware. The vulnerability analysis of the DJI Neo yielded a deterministic threat model, establishing that all severe exploits require either unencrypted management frames or unauthorized network access. Driven by these specific constraints, this research engineered a targeted, strictly rules-based IDRS.

Live-flight evaluations validated the operational capabilities of the custom architecture. The IDRS matched the 100% detection rate of industry-standard passive systems like Kismet while uniquely providing an active mitigation mechanism that autonomously severed malicious connections before internal exploits could begin. Because the system maintained a stable, low-overhead CPU footprint optimized for battery-constrained embedded environments, this relatively cheap hardware setup offers a scalable blueprint for securing commercial drone operations against wireless cyber-physical attacks.

Acknowledgments

The authors thank Jad Zarzour of Mason’s Innovation Lab for assistance with experimental setup and tooling. The authors used AI-assisted tools (OpenAI ChatGPT) for language refinement during the preparation of this manuscript. All technical content, analyses, and conclusions are the original work of the authors.

References

- [1] Shakhathreh, H., Sawalmeh, A. H., Al-Fuqaha, A., Dou, Z., Almaita, E., Khalil, I., Othman, N. S., Khreishah, A., and Guizani, M., “Unmanned Aerial Vehicles (UAVs): A Survey on Civil Applications and Key Research Challenges,” *IEEE Access*, Vol. 7, 2019, pp. 48572–48634. <https://doi.org/10.1109/ACCESS.2019.2909530>.
- [2] Roy, S., Sankaran, S., and Zeng, M., “Green Intrusion Detection Systems: A Comprehensive Review and Directions,” *Sensors*, Vol. 24, No. 17, 2024, p. 5516.
- [3] Yaacoub, J.-P., Noura, H., Salman, O., and Chehab, A., “Security analysis of drones systems: Attacks, limitations, and recommendations,” *Internet of Things*, Vol. 11, 2020, p. 100218. <https://doi.org/10.1016/j.iot.2020.100218>.
- [4] Abbas Shah, S. F., Mazhar, T., Shloul, T. A., Shahzad, T., Yu-Chen, H., Mallek, F., and Hamam, H., “Applications, challenges, and solutions of unmanned aerial vehicles in smart city using blockchain,” *PeerJ Computer Science*, 2024.
- [5] Mustafa Abro, E., Ghulam, Saiful Azrin, B. M. Z., Rana, J. M., Asirvadam, V. S., and Laouti, A., “Comprehensive Review of UAV Detection, Security, and Communication Advancements to Prevent Threats,” *Drones*, Vol. 6, No. 10, 2022, p. 284.
- [6] Wallace, R. J., Lee, S.-A., Rice, S., Winter, S. R., Gomez, L. M., Arboleda, G., Terwilliger, B., Haritos, T., Silas, K., and Snyder, P., “Drone Traffic Analysis (A11L.UAS.122) Annual Report (2025),” Tech. Rep. A11L.UAS.122–A83, Federal Aviation Administration, ASSURE UAS Center of Excellence, Washington, DC, January 2026. Technical Report.
- [7] Forum of Incident Response and Security Teams (FIRST), “Common Vulnerability Scoring System v4.0 Specification Document,” <https://www.first.org/cvss/specification-document>, November 2023. Accessed: 2026-03-08.
- [8] Kershaw, M., “Kismet Wireless Intrusion Detection System,” <https://www.kismetwireless.net/>, 2024. Accessed: 2026-03-08.
- [9] Bayrak, S., “Unveiling intrusions: explainable SVM approaches for addressing encrypted Wi-Fi traffic in UAV networks,” *Knowledge and Information Systems*, Vol. 66, No. 11, 2024, pp. 6675–6695.
- [10] Westerlund, O., and Asif, R., “Drone Hacking with Raspberry-Pi 3 and WiFi Pineapple: Security and Privacy Threats for the Internet-of-Things,” *2019 1st International Conference on Unmanned Vehicle Systems-Oman (UVS)*, 2019, pp. 1–10. <https://doi.org/10.1109/UVS.2019.8658279>.

- [11] Mekdad, Y., Aris, A., Babun, L., Fergougui, A. E., Conti, M., Lazzeretti, R., and Uluagac, A. S., “A survey on security and privacy issues of UAVs,” *Computer Networks*, Vol. 224, 2023, p. 109626. <https://doi.org/https://doi.org/10.1016/j.comnet.2023.109626>.
- [12] Pratama, D., Moon, J., Ari Laksmono, A. M., Yun, D., Iqbal, M., Jeong, B., Ji, J. H., and Kim, H., “Behind The Wings: The Case of Reverse Engineering and Drone Hijacking in DJI Enhanced Wi-Fi Protocol,” *2024 International Conference on Platform Technology and Service (PlatCon)*, 2024, pp. 127–132. <https://doi.org/10.1109/PlatCon63925.2024.10830741>.
- [13] Schiller, N., Chlosta, M., Schloegel, M., Bars, N., Eisenhofer, T., Scharnowski, T., Domke, F., Schönherr, L., and Holz, T., “Drone Security and the Mysterious Case of DJI’s DroneID,” *Proceedings of the 2023 Network and Distributed System Security Symposium (NDSS)*, 2023. <https://doi.org/10.14722/ndss.2023.24217>.
- [14] Mitchell, R., and Chen, I.-R., “Adaptive Intrusion Detection of Malicious Unmanned Air Vehicles Using Behavior Rule Specifications,” *IEEE transactions on systems, man, and cybernetics. Systems*, Vol. 44, No. 5, 2014, pp. 593–604.
- [15] Asif, M., Rahman, M. A., Akkaya, K., Shahriar, H., and Cuzzocrea, A., “Adversarial Data-Augmented Resilient Intrusion Detection System for Unmanned Aerial Vehicles,” *2023 IEEE International Conference on Big Data (BigData)*, 2023, pp. 5428–5437. <https://doi.org/10.1109/BigData59044.2023.10386140>.
- [16] Hassler, S. C., Mughal, U. A., and Ismail, M., “Cyber-Physical Intrusion Detection System for Unmanned Aerial Vehicles,” *IEEE Transactions on Intelligent Transportation Systems*, Vol. 25, No. 6, 2024, pp. 6106–6117. <https://doi.org/10.1109/TITS.2023.3339728>.
- [17] Kulp, P., and Mei, N., “A Framework for Sensing Radio Frequency Spectrum Attacks on Medical Delivery Drones,” *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2020, pp. 408–413. <https://doi.org/10.1109/SMC42975.2020.9283478>.
- [18] IEEE, “IEEE Standard for Information technology - Telecommunications and information exchange between systems - Local and metropolitan area networks - Specific requirements. Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications Amendment 4: Protected Management Frames,” *IEEE Std 802.11w-2009 (Amendment to IEEE Std 802.11-2007 as amended by IEEE Std 802.11k-2008, IEEE Std 802.11r-2008, and IEEE Std 802.11y-2008)*, 2009, pp. 1–111. <https://doi.org/10.1109/IEEESTD.2009.5278657>.
- [19] Wright, J., and Cache, J., *Hacking Exposed Wireless: Wireless Security Secrets & Solutions*, 3rd ed., McGraw-Hill Education, New York, 2015.
- [20] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., and Duchesnay, E., “Scikit-learn: Machine Learning in Python,” *J. Mach. Learn. Res.*, Vol. 12, No. null, 2011, p. 2825–2830.
- [21] Wireshark Foundation, “Wireshark Network Protocol Analyzer,” <https://www.wireshark.org/>, 2024. Accessed: 2026-03-08.
- [22] d’Otreppe, T., “Aircrack-ng: An 802.11 WEP and WPA-PSK keys cracking program,” <https://www.aircrack-ng.org/>, 2024. Accessed: 2026-03-08.